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Roll No

EC-7004(2)-CBGS

B.E. VII Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Power Electronics

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions
ii) All questions carry equal marks.

1. a) Discuss the transistor model of GTO and derive the expression for anode current in terms of current gain.
b) Draw and explain the static characteristics of a Triac.
2. Explain two transistor analogy for a thyristor with the help of neat illustrations. On what factors does the $\frac{di}{dt}$ rating of a thyristor depend.
3. a) Explain the operation of single-phase fully controlled bridge rectifier for RL load and derive the expression for Ripple factor.
b) A single-phase half controlled bridge rectifier supplied from 240 V, 50 Hz AC source feeds a resistance load of 50Ω . Calculate the power dissipated in the load when the rectifier operates at a firing angle of 45° .

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4.
 - a) Explain the single-phase dual converter with proper circuit diagram and different waveforms.
 - b) Draw the circuit diagram of three-phase, six-pulse controlled rectifier. Find out the average and rms values of voltage and current for resistive load.
5.
 - a) Explain the switching behaviour of GTO. How does a GTO differ from a conventional thyristor.
 - b) A single-phase semi-converter feeds a load comprising of a resistance of 10Ω and a large inductance to provide constant and ripple free load current. Calculate the input and output performance parameters of the converter circuit if firing angle delay is 45° and input A.C. voltage is 12V, 50Hz. Neglect device and other drops.
6.
 - a) Explain how phase-controlled rectifiers reduce the supply line power factor. Also discuss the methods to improve power factor.
 - b) A single-phase AC voltage controller comprising a thyristor and diode supplies purely resistive load of 10Ω from a 230V, 50Hz supply. Calculate
 - i) The r.m.s value of load voltage
 - ii) Input power factor
7.
 - a) A single-phase full bridge inverter with an RL load of $R=20\Omega$ and $L=10\text{mH}$, produces a square wave. It is fed from a 100 V dc input. Find the rms load voltage, first fundamental rms and total harmonic distortion.
 - b) What are the advantages of current-driven inverters over voltage-driven inverters? Describe the working of a single-phase current driven inverter.
8. Write short note on any two of the following:
 - a) Morgan Chopper
 - b) McMurray bedford inverter
 - c) Buck boost regulator

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